Network Value and the Metcalfe Utility Function:

Estimating the Total Assets of Meta Platforms as a Function of Monthly Active Users

Course: Machine Learning
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1. Problem Statement

This project studies how the value of Meta Platforms changes as the number of users increases. The number of users is measured using Monthly Active Users (MAU), while the company's value is measured using its total assets from annual reports between 2008 and 2023. The project compares two mathematical models to see which better predicts Meta's total assets based on the number of users: the Metcalfe utility function, which assumes network value grows through connections between users, and a simple linear regression model. The project also examines how the chosen model relates to the concept of diminishing marginal utility, meaning whether each additional user adds less value to the network over time.

1.1 Research Questions

This paper addresses the following two research questions:

(a) Which model function provides a better estimate of Meta's total assets as a function of monthly active users: the Metcalfe utility function or a linear regression function?

(b) How does the preferred model relate to the classical law of diminishing marginal utility, and what are the economic implications of this relationship?

2. Methodology

This section presents the methodological framework adopted for the analysis. It describes the data source and modelling approaches applied to estimate Meta's total assets as a function of Monthly Active Users (MAU). Furthermore, the mathematical formulation of the Linear Regression and Metcalfe utility models, together with the evaluation methodology based on the coefficient of determination (R^2), are introduced.

2.1 Linear Regression Model

The first modelling approach applied in this study is Linear Regression. The model assumes that Meta's total assets increase proportionally with the number of Monthly Active Users (MAU), implying that each additional user contributes a constant amount to the overall network value. The Linear Regression model is expressed as:

$$\mathbf{y = b_0 + b_1 x}$$

Where:
(x) represents Monthly Active Users (MAU)
(y) represents Total Assets
(b_0) is the intercept term
(b_1) is the slope coefficient

The model parameters were estimated using the least-squares regression method implemented in Python using the Scikit-learn library (LinearRegression class from sklearn.linear_model) (*Pedregosa et al., 2011)* . The intercept and slope values were obtained using the model attributes **intercept_** and **coef_** after fitting the model to the dataset.
y = - 47027.98 + 72.53x

The estimated parameters are:
b_0 = - 47027.98 , which represents the intercept of the model, and
b_1 = 72.53 , which represents the slope coefficient.

The positive value of the slope indicates that the total assets increase with the growth of Monthly Active Users. Under the linear assumption, every additional unit increase in MAU contributes approximately **72.53 units** i.e approximately 72.53 million USD to Meta's total assets.

The accuracy or goodness of fit of the model was subsequently evaluated using the coefficient of determination R^2, which measures how well the regression model explains the observed variation in the data.

$$R^2 = 1 - \frac{\Sigma(y_i - \hat{y}_i)^2}{\Sigma(y_i - \bar{y})^2}$$

Where,
(y_i) represents the observed values
($\hat{y}_i$) represents the predicted values
($\bar{y}$) the mean of the observed values
(R^2) coefficient of determination (i.e How much of the variation in the observed data is explained by the model.)

The value of (R^2) was calculated in Python using the Scikit-learn metrics library through the **r2_score()** function *(Pedregosa et al., 2011)* , which compares the observed asset values with the predicted values generated by the regression model. The Linear Regression model produced:
R^2 = 0.898

This indicates that approximately **89.8%** of the variation in Meta's total assets can be explained by the linear relationship with Monthly Active Users (MAU).

2.2 Metcalfe Utility Function

The second modelling approach applied in this study is the Metcalfe Utility Function. Unlike Linear Regression, the Metcalfe model assumes that the value of a network increases non-linearly with the number of connected users due to network effects. The model is based on Metcalfe's Law, which states that the utility or value of a network is proportional to the square of the number of users. The Metcalfe Utility Function is expressed as:

$$u_M(x) = a \cdot \frac{x(x-1)}{2}$$

Where:

- **x** represents the number of nodes in the network. In this study, the nodes are approximated by Monthly Active Users (MAU) of the Meta Platforms.
- $\mathbf{u_M(x)}$ represents the estimated network value (Total Assets)
- **'a'** represents scaling coefficient used to fit the model to the observed data
- $\frac{x(x-1)}{2}$ represents the total number of possible user-to-user network connections and forms the basis of the Metcalfe model.

The model parameter was estimated in Python by first calculating the Metcalfe term $\frac{x(x-1)}{2}$ using the NumPy library. The scaling coefficient 'a' was then determined using the least-squares estimation approach through the np.sum() function. NumPy was used for mathematical computations, while Pandas was used for data loading and preprocessing *(McKinney, 2010).*

$$a = \frac{\Sigma(y_i u_i)}{\Sigma u_i^2}$$

where:
y_i represents the observed asset values
u_i represents the calculated Metcalfe term

The scaling coefficient obtained from the implementation was:
a = 0.041922

Thus, the fitted Metcalfe model becomes:

$$u_M(x) = 0.041922 \ . \ \frac{x(x-1)}{2}$$

The positive value of the scaling coefficient indicates that Meta's total assets increase with network expansion and increasing user interactions. Since the model follows a quadratic relationship, the value grows faster as the number of users increases.

The accuracy or goodness of fit of the model was evaluated using the coefficient of determination $\mathbf{R^2}$:

$$R^2 = 1 - \frac{\Sigma(y_i - \hat{y}_i)^2}{\Sigma(y_i - \bar{y})^2}$$

Where,
(y_i) represents the observed values
($\hat{y}_i$) represents the predicted values
($\bar{y}$) the mean of the observed values
(R^2) coefficient of determination i.e How much of the variation in the observed data is explained by the model.

The function compares the observed asset values with the values predicted by the Metcalfe model and determines how much of the variation in the observed data is explained by the model. Values of $\mathbf{R^2}$ closer to 1 indicate a better fit and stronger explanatory capability. The Metcalfe model produced: $\mathbf{R^2 = 0.976}$

This indicates that approximately **97.6%** of the variation in Meta's total assets can be explained by the **Metcalfe Utility Function** relationship with Monthly Active Users (MAU).

3. Model Comparison

Which model function should be preferred to estimate the total assets as a function of the monthly active users ?

The below graph illustrates the observed values of Meta's total assets (blue scatter points) together with the fitted Linear Regression model (solid blue line) and Metcalfe model (orange curve) for the period

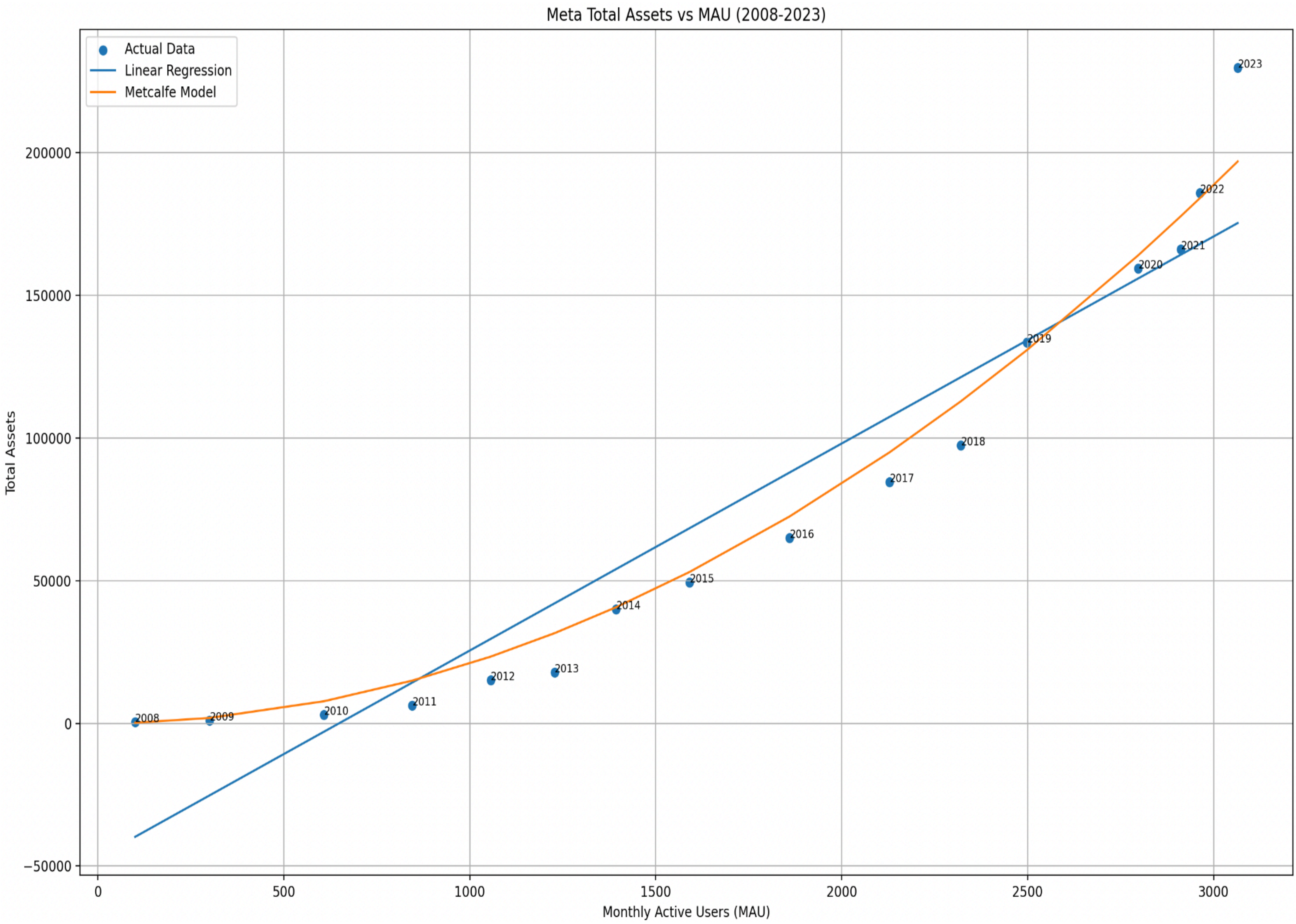
2008–2023. The figure shows that while the Linear Regression model captures the general upward trend, the orange Metcalfe curve follows the observed data points more closely, particularly during the later years characterized by rapid growth.

The Linear Regression model produced a coefficient of determination **R^2 = 0.898** , indicating that approximately **89.8%** of the variation in Meta's total assets can be explained by the linear relationship with Monthly Active Users.

The Metcalfe Utility Function produced **R^2 = 0.976** , indicating that approximately **97.6%** of the variation in Meta's total assets can be explained by the network-based Metcalfe relationship.

As, **0.976 > 0.898**

The Metcalfe Utility Function provides a better estimation of Meta's total assets. **Therefore, the preferred model for estimating Meta's total assets as a function of Monthly Active Users (MAU) is the Metcalfe Utility Function.**



4. Relationship of the Metcalfe utility function to the Law of Diminishing Marginal Utility

4.1 The Preferred Model and the Law of Diminishing Marginal Utility

Section 3 showed that the Metcalfe model fits Meta's asset data better than the linear model. This suggests that value creation on digital platforms is driven more by network effects than by a simple one-to-one increase in users *(Katz and Shapiro, 1985)* .

In classical microeconomics, diminishing marginal utility implies that each additional unit contributes less value than the previous one. While this logic is appropriate for private goods, it is less suitable for digital networks, where an additional user can increase value for many others.

When a new user joins Meta, the platform gains more than individual demand; it also gains additional possible interactions. As a result, digital networks may exhibit increasing marginal utility rather than diminishing marginal utility.

4.2 Increasing Marginal Utility Under the Metcalfe Model

The Metcalfe model captures this relationship by linking value to the number of possible user connections:

$$u_M(x) = a \cdot \frac{x(x-1)}{2}$$

Here, x denotes Monthly Active Users and a represents the scaling coefficient. Because the number of possible connections grows approximately with x^2, the value contributed by each additional user can increase as the network expands.

In practical terms, a new user adds more value to a large platform than to a smaller one.

4.3 Economic Implications for Digital Platform Economies

These findings have several implications for digital platform economies.

First, scale matters. Once a platform reaches critical mass, network effects can accelerate value creation, which helps explain the strong emphasis on user growth.

Second, network effects can reinforce market concentration. Larger platforms may therefore develop persistent advantages over smaller competitors.

Third, total assets remain an imperfect proxy for network value. They also reflect other factors, including investment and acquisitions.

Fourth, platform economics differs from standard marginal analysis. For network-based platforms, additional users may increase marginal value rather than reduce it.

4.4 Limitations and Open Questions

The Metcalfe model nevertheless has limitations. It assumes equal value across users and connections, even though real social networks are uneven and clustered *(Briscoe, Odlyzko and Tilly, 2006)*.

It also treats Meta as a single network, although Facebook, Instagram, and WhatsApp may differ in their patterns of interaction and connectivity.

Future research should examine whether this relationship weakens as user growth slows and whether sub-quadratic models provide a better fit for mature platforms.

5. Conclusion

The results showed that the Metcalfe Utility Function outperformed the Linear Regression model, achieving an $\mathbf{R^2}$ value of 0.976, compared to 0.898 for Linear Regression. The graphical analysis further confirmed that the Metcalfe model followed the observed asset values more closely, particularly during periods of rapid network growth.

These findings indicate that the value creation process in digital social networks is better explained by network effects rather than a simple linear relationship. The results further suggest that digital platforms such as Meta exhibit increasing marginal utility, which contrasts with the classical law of diminishing marginal utility. This study set out to determine which model better estimates Meta Platforms' total assets as a function of Monthly Active Users: a conventional Linear Regression or the Metcalfe Utility Function. The results clearly favour the Metcalfe model, which achieved a coefficient of determination of $R^2 = 0.976$ compared to $R^2 = 0.898$ for the linear alternative.

The preferred Metcalfe model implies increasing marginal utility, in contrast to the classical law of diminishing marginal utility. In digital social networks, each additional user generates positive network externalities by increasing value for other users, which the model's quadratic structure captures more effectively than a linear specification.

These findings suggest that classical microeconomic models require adaptation for network-based platforms. The Metcalfe framework provides a theoretically and empirically supported account of platform value, though future research should test whether this relationship weakens under slower user growth and whether sub-quadratic models better describe mature markets.

6. References

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7. Appendix

Python Code :

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score
```

```
#Load Data
df = pd.read_csv("Facebook.csv")
# Split single column into 3 columns
df = df.iloc[:,0].str.split("\t",expand=True)
# Rename columns
df.columns = ["Year","MAU","Total_Assets","Empty1","Empty2"]
# Keep only needed columns
df = df[["Year","MAU","Total_Assets"]]
# Convert to numeric
df = df.apply(pd.to_numeric)
print(df)
# ======================================
# Variables
# ======================================
x = df["MAU"].values
y = df["Total_Assets"].values
years = df["Year"].values
x_reshaped = x.reshape(-1,1)
# ======================================
# Linear Regression
# ======================================
linear_model = LinearRegression()
linear_model.fit(x_reshaped,y)
linear_predictions = linear_model.predict(x_reshaped)
#Get intercept and slope
intercept = linear_model.intercept_
slope = linear_model.coef_[0]
linear_r2 = r2_score(y,linear_predictions)
print("\nLinear Regression R² =",linear_r2)
# ======================================
# Metcalfe Model
# u(x)=a*x(x-1)/2
# ======================================
metcalfe_term = (x*(x-1))/2
a = np.sum(y*metcalfe_term)/np.sum(metcalfe_term**2)
metcalfe_predictions = (a*metcalfe_term)
metcalfe_r2 = r2_score(y,metcalfe_predictions)
print("Metcalfe R² =",metcalfe_r2)
# ======================================
# Plot
# ======================================
plt.figure(figsize=(10,6))
# Actual data
plt.scatter(x,y,label="Actual Data")

# Add year labels
for i in range(len(years)):
   plt.annotate(years[i],(x[i],y[i]),fontsize=8 )
# Linear Regression
plt.plot(x,linear_predictions,label="Linear Regression")
# Metcalfe
plt.plot(x,metcalfe_predictions,label="Metcalfe Model")
plt.xlabel("Monthly Active Users (MAU)")
plt.ylabel("Total Assets")
plt.title("Meta Total Assets vs MAU (2008-2023)")
plt.grid(True)
plt.legend()
plt.show()
```